



Ophthalmology Mnemonics

Compiled Reference

Parts I, II & III · 71 High-Yield Exam Mnemonics

For ICO / FAICO / PD-CET / FRCOphth / OF / MAICO/OKAT / DNB / MS / DO exam preparation

Mnemonics for cornea, glaucoma, retina, neuro-ophthalmology, orbit, oculoplastics, and strabismus — shuffled, one per page.



OPHTHA MCQ APP

ONE APP FOR ALL OPHTHALMOLOGY EXAMS
ICO | FAICO | PD-CET | FRCOphth | OF | MAICO/OKAT | DNB | MS | DO
BUILT BY OPHTHALMOLOGISTS, FOR OPHTHALMOLOGISTS

Contents

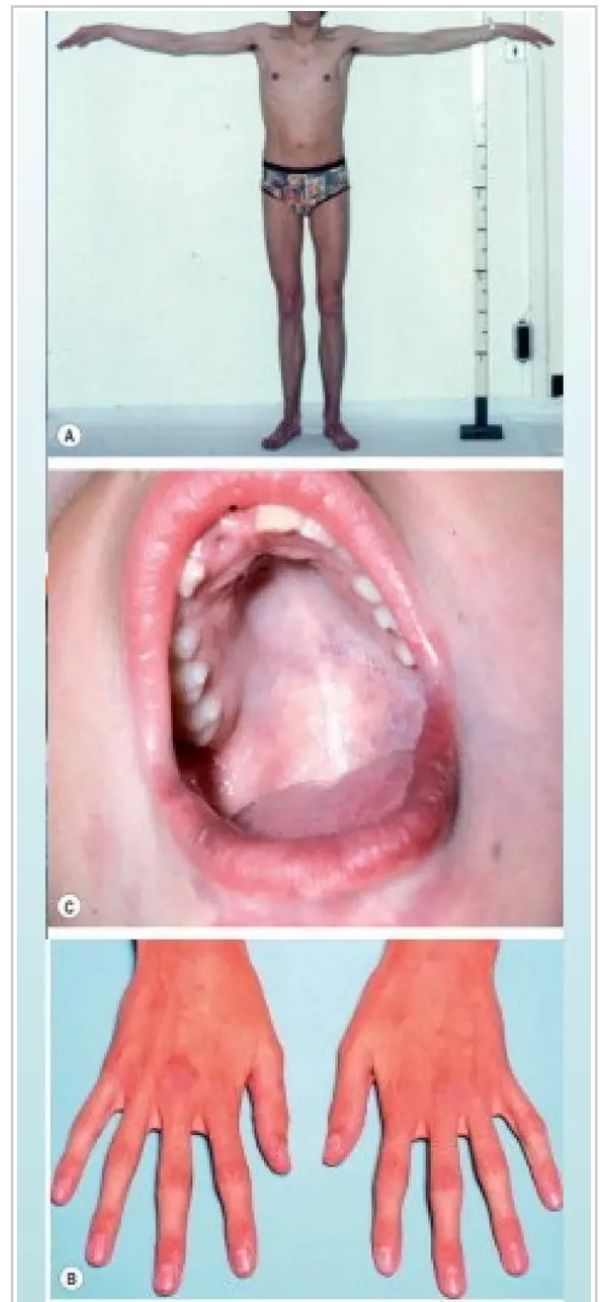
Order below is shuffled for review; each entry is tagged with its source part.

1. [I] Systemic features of Marfan syndrome
2. [I] Causes of choroidal neovascular membrane
3. [III] Systemic associations of keratoconus
4. [I] Microtropia
5. [II] Modified WHO grading of trachoma
6. [I] Interstitial keratitis causes
7. [II] Muscle involvement in TED
8. [II] Differentiating a naevus from a malignant melanoma
9. [I] Organisms invading healthy corneal epithelium
10. [II] Causes of lid retraction
11. [II] Buphthalmos
12. [III] Systemic complications of steroid therapy
13. [I] Features of retinal detachment
14. [I] Uniocular diplopia
15. [I] Causes of cherry red spot on retina
16. [II] Ocular features of Marfan syndrome
17. [II] Duane retraction syndrome
18. [II] Angioid streaks
19. [III] Systemic associations of RP — Bassen-Kornzweig syndrome
20. [II] Causes of papilledema
21. [II] Differential diagnosis of leukocoria
22. [II] Drugs causing nystagmus
23. [II] Causes of pseudotumor cerebri
24. [I] Posterior scleritis
25. [III] Signs of congenital glaucoma
26. [I] Iridocorneal Endothelial Syndrome
27. [I] Actions of superior oblique muscle
28. [II] Color blindness
29. [II] Non-infectious vs suppurative corneal infiltrates
30. [III] Ocular features of acromegaly
31. [III] Viral retinitis
32. [I] Blue sclera
33. [III] Cryotherapy
34. [I] Congenital cloudy cornea
35. [III] Treatment of Acanthamoeba keratitis
36. [III] Therapeutic indications for contact lens
37. [I] Causes of trabecular pigmentation
38. [II] Neurofibromatosis I — ocular features
39. [I] Nerve palsy — causes
40. [III] Systemic associations of RP — Refsum's syndrome
41. [I] Vortex keratopathy associations
42. [I] Nerves passing through the superior orbital fissure
43. [III] Drugs to avoid in myasthenia gravis
44. [III] Causes of optic neuropathy
45. [III] Suggestive features of iris malignant melanoma
46. [III] Alport's syndrome
47. [I] Weill-Marchesani syndrome features
48. [II] Thyroid eye disease surgery
49. [II] Before dilatation of the pupil, don't forget
50. [II] Trachoma control
51. [II] Behcet's disease
52. [I] Neovascular Glaucoma — treatment
53. [III] Iris nodules in anterior uveitis
54. [III] Ocular pathological effects of pregnancy
55. [II] A and V pattern treatment
56. [II] Assessment of severity of thyroid eye disease
57. [III] Goldmann three-mirror lens
58. [III] Suspicious eyelid nevus
59. [I] Stromal corneal dystrophies
60. [I] Angle structures
61. [I] Keratoconus
62. [II] Whitnall's tubercle attachments
63. [II] A and V patterns
64. [I] Etiology of Neovascular Glaucoma
65. [I] Argyll Robertson pupil
66. [I] Indications of vitrectomy
67. [II] Neuroectoderm derivatives
68. [II] Causes of proptosis
69. [II] Crossed-uncrossed diplopia
70. [I] Systemic associations of keratoconus
71. [I] Drugs causing cataract

Systemic features of Marfan syndrome

MARFANS

- M**itral prolapse
- A**ortic dissection
- R**egurgitant aortic valve
- F**ingers long (arachnodactyly)
- A**rm span > height
- N**asal voice (high arched palate)
- S**ternal excavation



Causes of choroidal neovascular membrane

HAMMAR

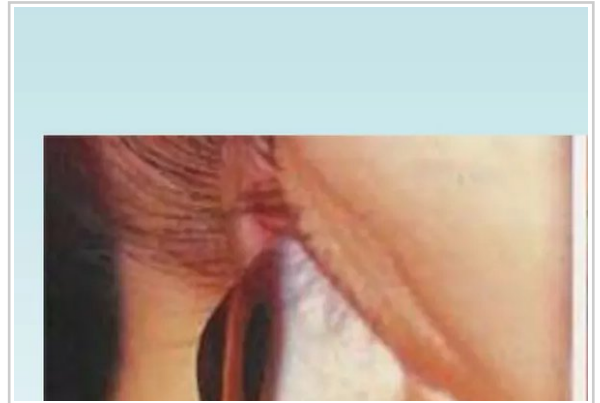
- H**istoplasmosis
- A**RMD
- M**ultifocal Choroiditis
- M**yopia
- A**ngioid streaks
- R**upture of the choroid



Systemic associations of keratoconus

5 C'S

- C**hromosomal disorder (Down's syndrome)
- C**onnective tissue disorder (Marfan's syndrome, osteogenesis imperfecta)
- C**utaneous disorders (atopic dermatitis)
- C**ongenital ocular anomalies (aniridia, Leber's congenital amaurosis)
- C**ontact lens wearer



Microtropia

3 A'S

Anisometropia

Angle small

Absent central field (Central suppression scotoma)

Modified WHO grading of trachoma

FISTO

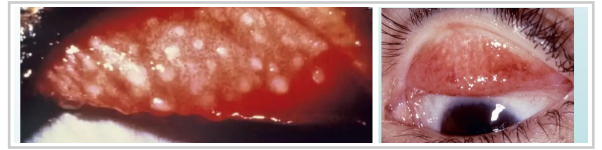
TF = trachoma **f**ollicles

TI = trachomatous **i**nflammation

TS = trachomatous conjunctival **S**carring

TT = trachomatous **t**richiasis

CO = Corneal **O**paCity



Interstitial keratitis causes

TIC TAC'S

- T**uberculosis
- I**nherited syphilis (Congenital syphilis)
- T**rypanosomiasis
- A**cquired syphilis
- C**ogan's syndrome
- S**arcoidosis



Muscle involvement in TED

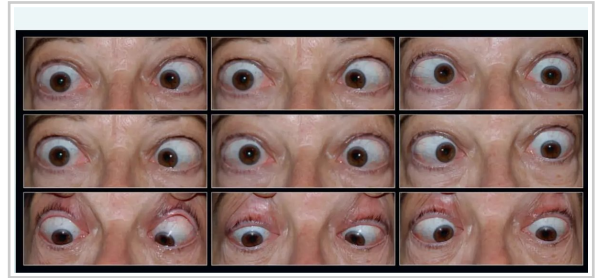
I MUST SLEEP LATE

Inferior rectus

Medial Rectus

Superior rectus

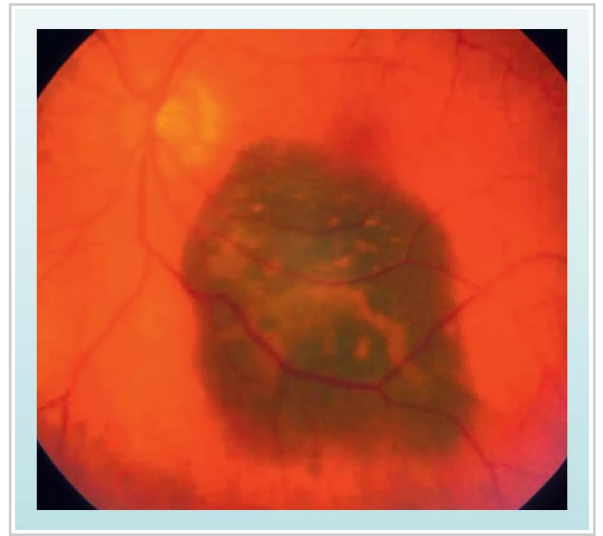
Lateral rectus



Differentiating a naevus from a malignant melanoma

TO FIND SMALL OCULAR MELANOMA

- T**hickness (> 2mm)
- F**luid (subretinal)
- S**ymptoms
- O**range pigments
- M**argin touching disc



Organisms invading healthy corneal epithelium

NDLH

Nisseria

Diphtheria

Listeria

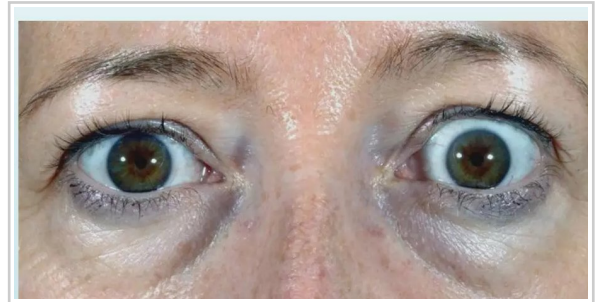
Haemophilus



Causes of lid retraction

4M 4P

- M**yasthenia
- M**yotonia Dystrophica
- M**arcus-Gunn jaw winking
- M**etabolic (Uremia)
- P**arinaud syndrome
- P**arkinsonism
- P**tosis (opposite eye)
- P**alsy (3rd nerve apparent regeneration)



Buphthalmos

5 B'S

Boys

Bilateral in 2/3

Blepharospasm

Blue sclera

Bull's eye (big eyes)



Systemic complications of steroid therapy

CUSHINGS

- C**ardiac complications (arrhythmias, heart failure)
- U**lcer (gastric ulcer)
- S**uppression of hypothalamic-pituitary-adrenal axis (shock)
- H**ypertension, **H**irsutism, **H**epatic (liver failure)
- I**schemic necrosis of femur and osteoporosis
- N**eutropenia and infection
- G**rowth problems in children
- S** for psychosis



Features of retinal detachment

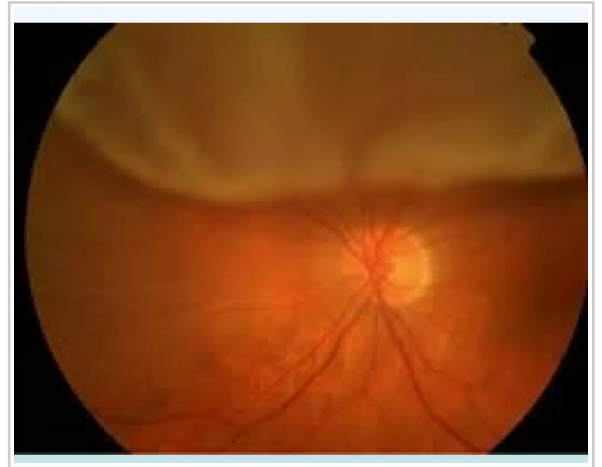
4 F'S

Floater

Flashes

Field loss

Falling visual acuity



Unocular diplopia

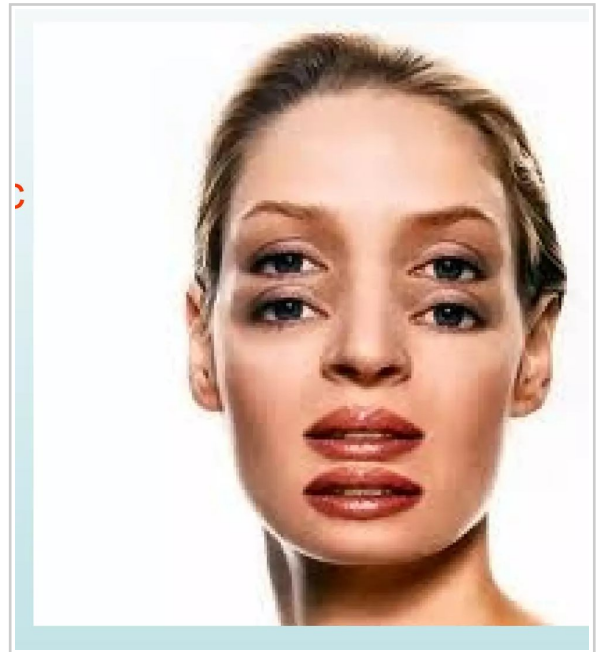
ABCD

Astigmatism

Behavioral: psychogenic

Cataract

Dislocated lens / iridoDialysis



Causes of cherry red spot on retina

CHERRY TREES NEVER GROW TALL

Central retinal artery occlusion

Tay-Sachs disease

Niemann-Pick disease

Gaucher's disease

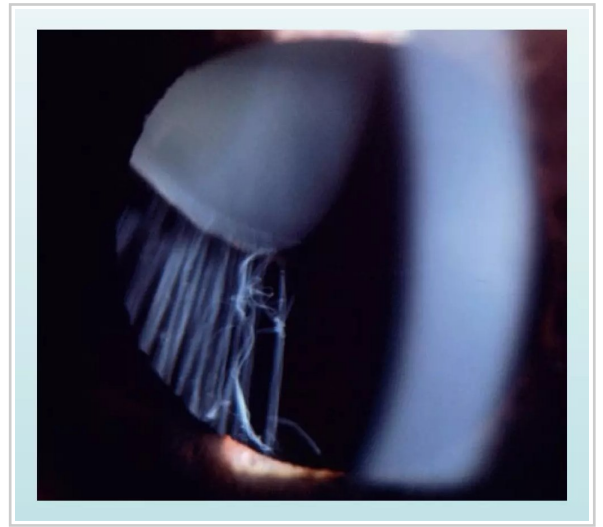
Trauma (Berlin's edema)



Ocular features of Marfan syndrome

CLUMPS

- C**upping (glaucoma)
- L**attice
- U**pward lens subluxation
- M**yopia
- Cornea **P**lana
- S**clera blue



Duane retraction syndrome

NUMBER OF D'S = SYNDROME NUMBER

Type 1 : Limited ab**D**uction

Type 2 : Limited a**DD**uction

Type 3 : Limited ab**D**uction & a**DD**uction



Angioid streaks

PEPSI

Ppseudoxanthoma elasticum (PXE)

Ehlers-Danlos syndrome

Paget's disease

Sickle cell anemia

Idiopathic (50%)



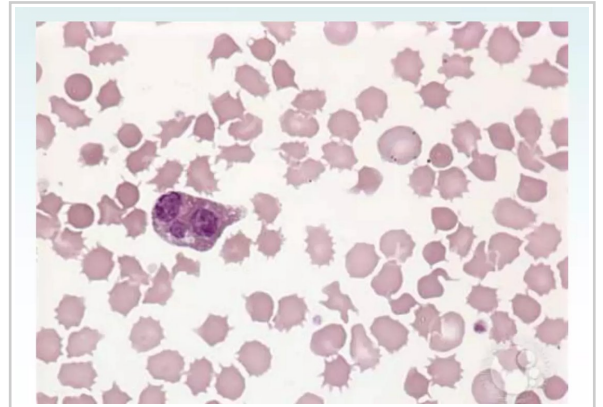
Systemic associations of RP — Bassen-Kornzweig syndrome

3 A'S

Ataxia

Acanthocytosis (red blood cell abnormality)

Abetalipoproteinemia (fat malabsorption)



Causes of papilledema

MOVIES / U DEVILS

Mass lesions, meningitis

Obstructive hydrocephalus

Venous obstructive disease (sinus thrombosis, dural shunts, jugular vein occlusion, SVC syndrome, right heart failure) and vascular disorders

Infection (abscess, meningitis)

Essential or other hypertension leading to malignant hypertension

Secondary causes of pseudotumor-cerebri-like syndrome (see U DEVILS)

—

Uremia

Drugs (tetracycline, lithium, acutane, nalidixic acid, retinoids, cyclosporin, amiodarone, growth hormone, norplant, sulfa, vitamin A)

Endocrine (hypoparathyroidism, Addison disease)

Vitamin A overuse

Iron deficiency anemia

Last menstrual period (pregnancy, birth control, menstrual irregularities)

Sleep apnea, steroid withdrawal



Differential diagnosis of leukocoria

PREDICT

Persistent hyperplastic primary vitreous
Retinoblastoma / **R**etinopathy of prematurity
Endophthalmitis
Dysplasia of the retina
Inflammatory cyclitic membrane
Congenital cataract / **C**oat's Disease
Toxocariasis



Drugs causing nystagmus

ABCD

Alcohol

Barbiturates

Carbamazepine

Diazepam



Causes of pseudotumor cerebri

IDIOPATHIC IDEA

Idiopathic

Infections — otitis media, mastoiditis, viral infections, etc.

Drugs — steroid withdrawal, Vitamin A intoxication, nalidixic acid, amiodarone, cyclosporin, minocycline

Endocrine — obese, amenorrheic woman of child-bearing age, hypoparathyroidism

Anemia

Posterior scleritis

POST SCLER

Proptosis

Ophthalmoplegia

Swelling of disc

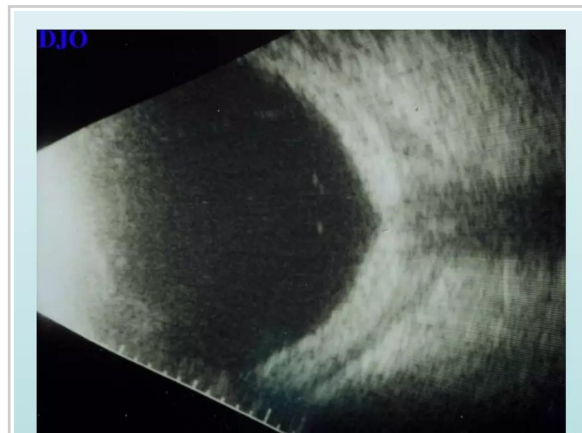
Thickening of sclera (US/CT) & T sign (fluid in sub-Tenon's space)

Subretinal exudates

Choroidal fo**L**ds

Exudative RD

Ring choroidal detachment



Signs of congenital glaucoma

ABCDDE

Axial myopia

Buphthalmos

Cloudy cornea

Descemet's breaks (Haab's striae)

Diameter of cornea enlarged

Disc cupping

Examination under GA



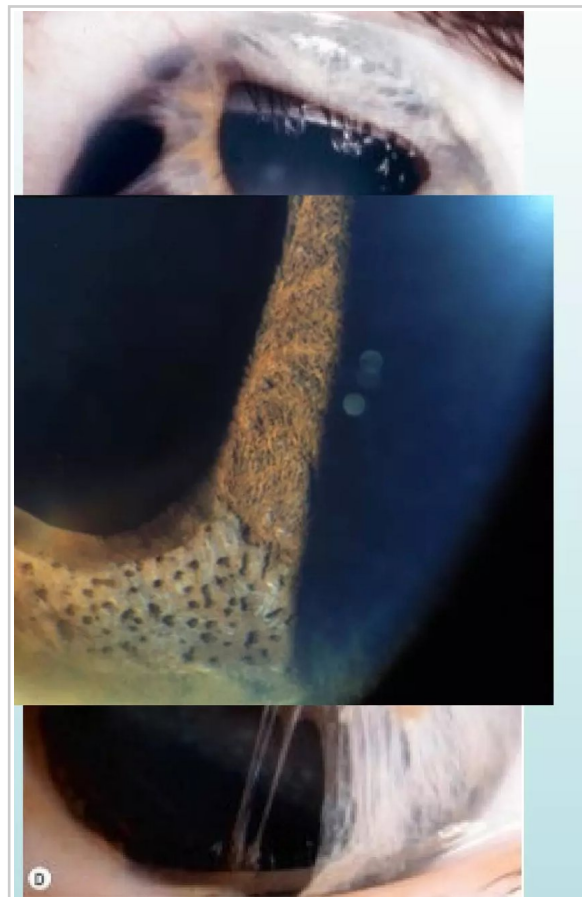
Iridocorneal Endothelial Syndrome

ICE

Iris Naevus

Chandler Syndrome

Essential Iris Atrophy



Actions of superior oblique muscle

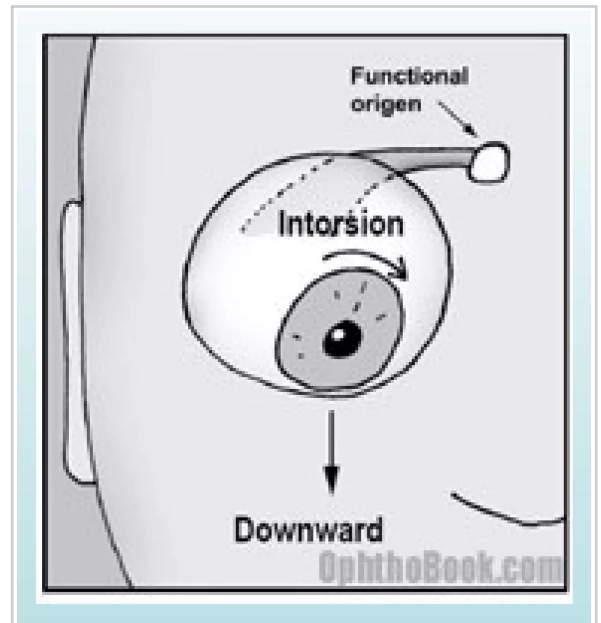
SOLID

Superior **O**blique

Lateral rotation

Intorsion

Depression



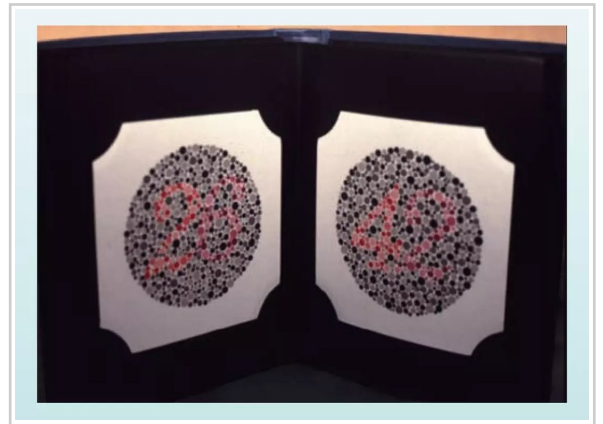
Color blindness

BUT GOD REPLY

Blue : **T**ritanopia

Green : **D**eutanopia

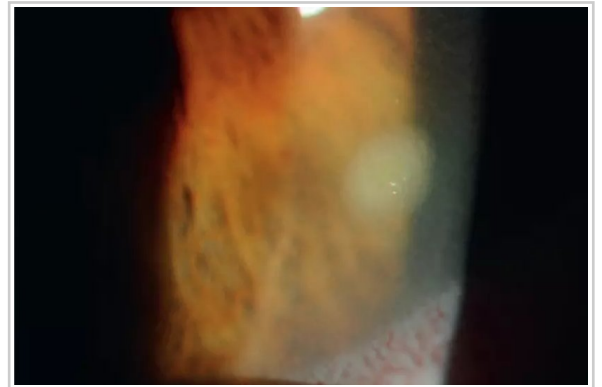
Red : **P**rotanopia



Non-infectious vs suppurative corneal infiltrates

PEDAL

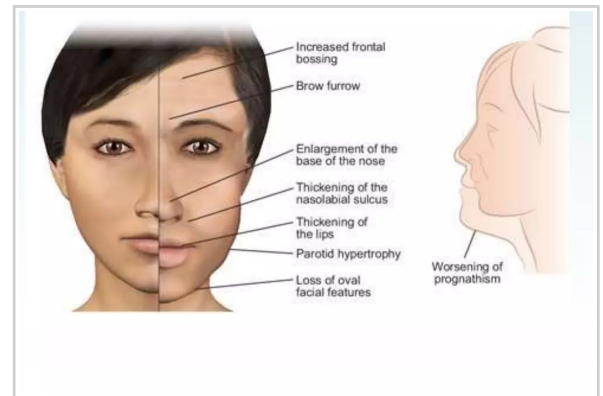
- P**ain
- E**pithelial defects (large > 1mm)
- D**ischarge (purulent)
- A**nterior chamber reaction (uveitis, hypopyon)
- L**ocation (central)



Ocular features of acromegaly

ACROM

- A**ngioid streaks
- C**hiasma syndrome
- R**etinopathy (DM and hypertensive retinopathy)
- O**ptic atrophy, papilledema
- M**uscle enlargement



Viral retinitis

ARN / PORN

Acute retinal necrosis (ARN) : **A**nterior

Progressive outer retinal necrosis (PORN) : **P**osterior



Blue sclera

HOPES BELIEF

- H**igh myopia
- O**steogenesis imperfecta
- P**seudoxanthoma elasticum
- E**hlers-Danlos syndrome
- S**cleritis
- B**uphthalmos



Cryotherapy

ABCD

Adhesion (retinal breaks)

Blood vessels (PDR new vessels, telangiectasia of Coat's disease)

Ciliary body (cyclodestructive procedures for glaucoma)

Cataract extraction (ICCE)

Destruction (lid and intraocular tumors, trichiasis)



Congenital cloudy cornea

STUMPED

- S**clerocornea
- T**ears in Descemet's membrane
- U**terine infections
- M**etabolic conditions
- P**eter's anomaly
- E**ndothelial dystrophy
- D**ermoid



Treatment of Acanthamoeba keratitis

ABCDE

- A**minoglycosides (neomycin)
- B**iguanides (polyhexamethylene biguanide)
- C**hlorhexidine
- D**iamidines (Brolene)
- E**conazole (Imidazoles)



Therapeutic indications for contact lens

5 P'S

- P**ain relief
- P**romote corneal healing
- P**rotect cornea
- P**erforated corneas
- P**harmaceutical delivery device



Causes of trabecular pigmentation

PIGMENT

Pseudoexfoliation & Pigment dispersion syndrome

Iritis

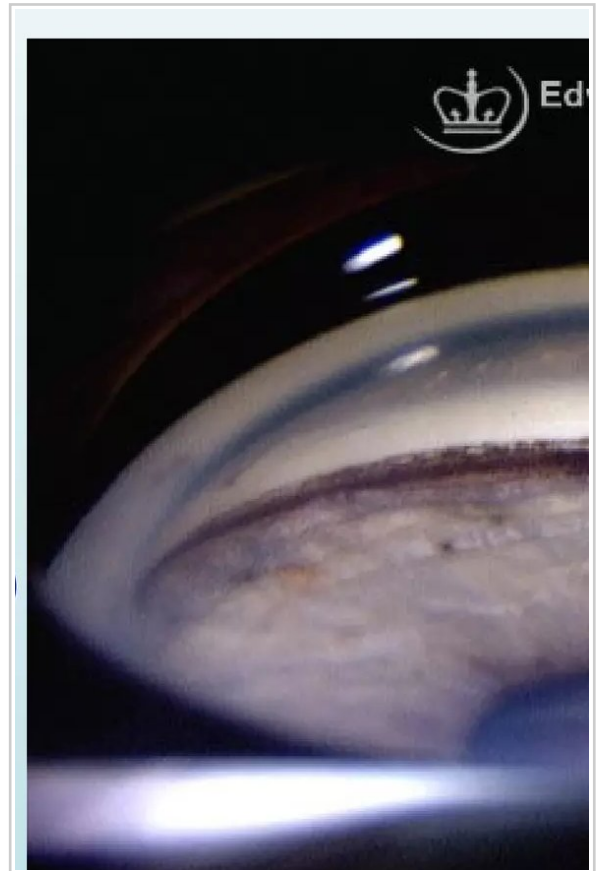
Glaucoma (Post angle closure Glaucoma)

Melanosis of angle (oculodermal melanosis)

Endocrine (Diabetes & Addison's Syndrome)

Naevus (Cogan-reese syndrome)

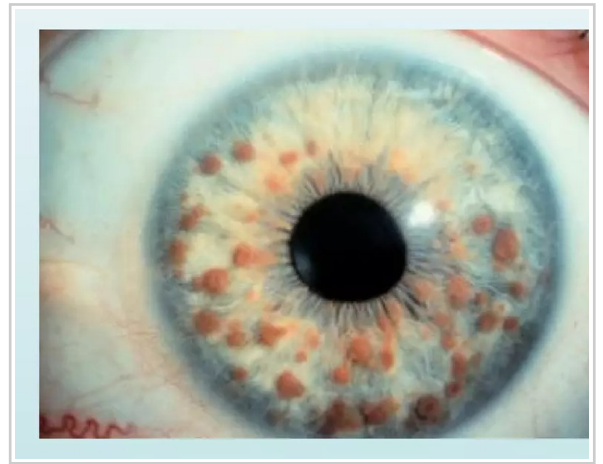
Trauma



Neurofibromatosis I — ocular features

OCULAR

- O**rbit : Optic Nerve Glioma
- C**ornea : Prominent corneal nerves
- U**vea : Lisch nodules
- L**id : Neurofibroma
- A**ngle : Glaucoma
- R**etina : Choroidal naevi & Astrocytoma



Nerve palsy — causes

VITAMIN

- Vascular
- Idiopathic
- Traumatic
- Aneurysm
- Miscellaneous
- Inflammatory
- Neoplasm



Systemic associations of RP — Refsum's syndrome

3 P'S

- P**hytanic acid metabolic defect
- P**eripheral neuropathy
- P**alpitations (cardiac arrhythmia)



Vortex keratopathy associations

ABCDEF

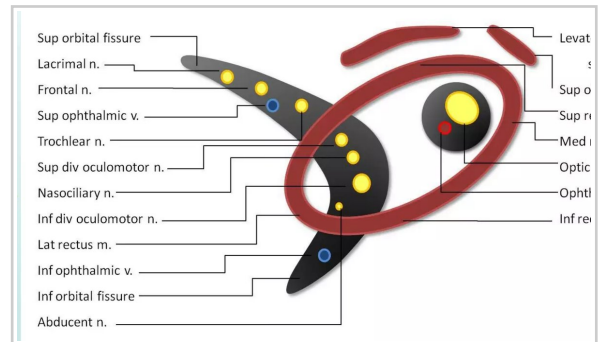
- A**rthritis (diclofenac)
- B**reast Cancer (Tamoxifen)
- C**ardiac (amiodarone)
- D**ementia / Depression (Chlorpromazine)
- E**nzyme deficiency (see F)
- F**abry's Disease — deficiency of the enzyme alpha galactosidase A



Nerves passing through the superior orbital fissure

LIVE FREE TO SEE NO INSULT AT ALL

- L**acrimal nerve
- F**rontal nerve
- T**rochlear nerve
- S**uperior division of 3rd nerve
- N**asociliary nerve
- I**nferior division of 3rd nerve
- A**bducens nerve



Drugs to avoid in myasthenia gravis

ABCD AND P

Aminoglycosides (gentamicin)

Neuromuscular **B**locker (curare, suxamethonium)

Chlorpromazine

Respiratory **D**epressants (morphine)

Procainamide

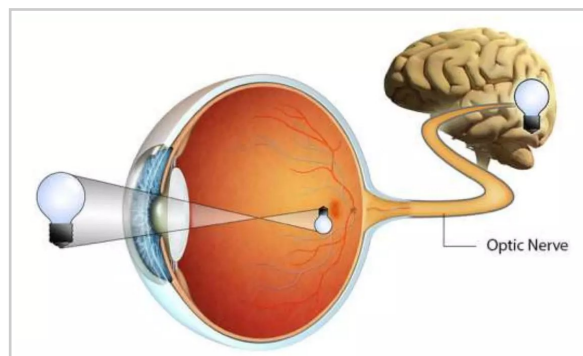
Penicillamine



Causes of optic neuropathy

NIGHT TICS

- N**euritis
- I**schemic
- G**ranulomatous
- H**ereditary
- T**raumatic
- T**oxic
- I**rradiation
- C**ompressive



Suggestive features of iris malignant melanoma

RIPPLE

Rubeosis

IOP increase

Pupil distortion

Photograph documentation of growth

Lens opacity

Ectropion uvea



Alport's syndrome

ALPORT + D

Anterior **L**enticonus

POsterior **P**Olymorphous corneal dystrophy

Retinal flecks / **R**enal failure

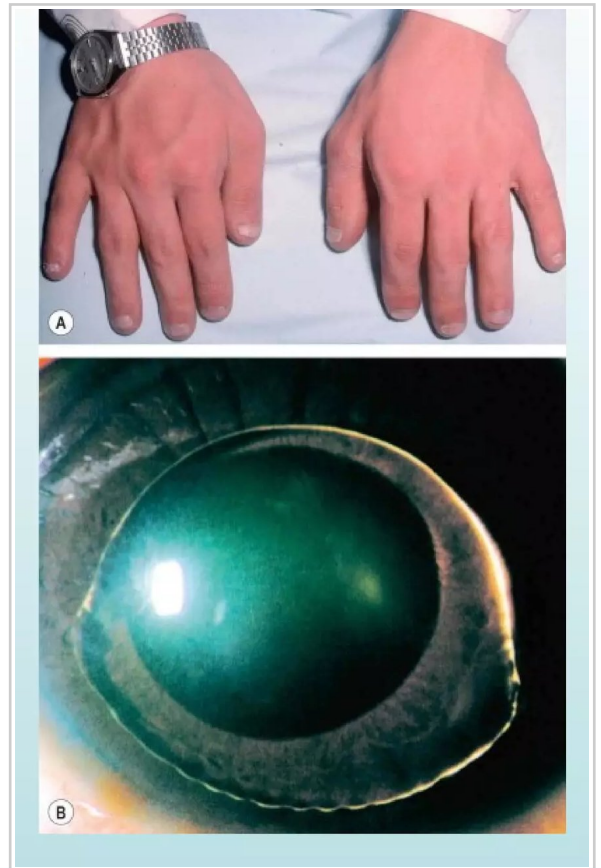
Deafness / XLD



Weill-Marchesani syndrome features

6 S'S

- S**hort
- S**tubby fingers
- S**ub-normal intelligence (in some cases)
- S**pherophakia
- S**ubluxed lens
- S**hallow AC



Thyroid eye disease surgery

OSQE

Orbital surgery

SQuint surgery

Eyelid surgery



Before dilatation of the pupil, don't forget

PART B

Pupillary reflex

Angle (of AC / of squint)

Rubeosis

Tension

BCVA



Trachoma control

SAFE

Surgery for trachomatous trichiasis

Antibiotic for active disease

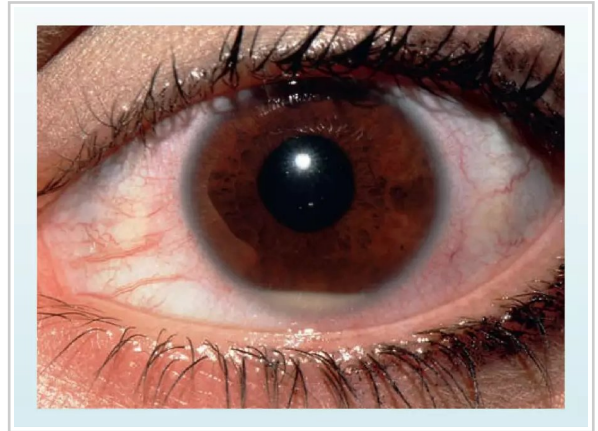
Face washing

Environmental improvements

Behcet's disease

ORAL UPSET

- Occlusive periphlebitis
- Retinitis
- Anterior uveitis
- Leakage from retinal vessels
- Ulceration (aphthous / genital)
- Pustules after skin trauma (Pathergy test)
- Scratching leaves lines (dermatographism)
- Erythema nodosum
- Thrombophlebitis



Neovascular Glaucoma — treatment

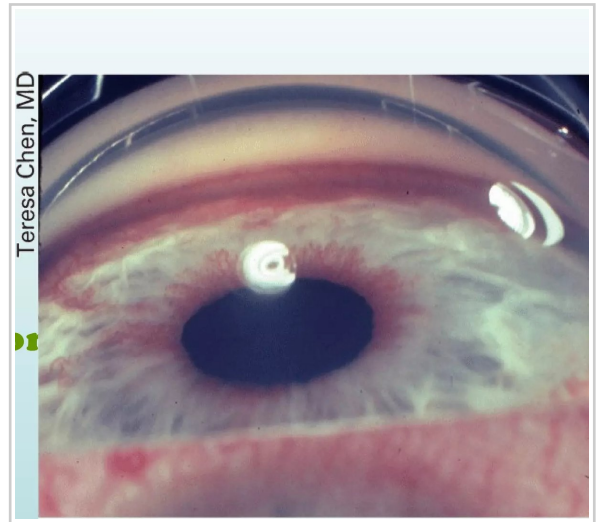
ABCD

Atropine

B-blockers

Cidamex

Dexamethasone



Iris nodules in anterior uveitis

KOEPPE / BUSACCA

Koepppe nodules : **p**upillary border

Busacca nodules : the iris stroma (**B**ase)



Ocular pathological effects of pregnancy

5 C'S

CSR

CRVO, CRAO, hypertensive retinopathy

Cortical blindness

Pseudotumor Cerebri

Coagulation disorders

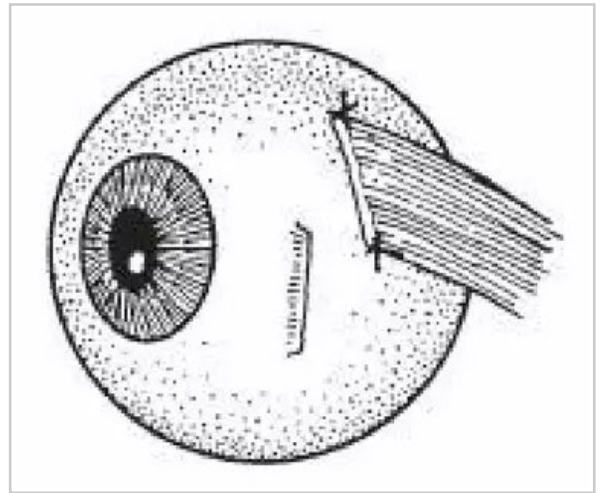


A and V pattern treatment

MALE

Medial rectus towards the **A**pex

Lateral Rectus towards the **E**nd



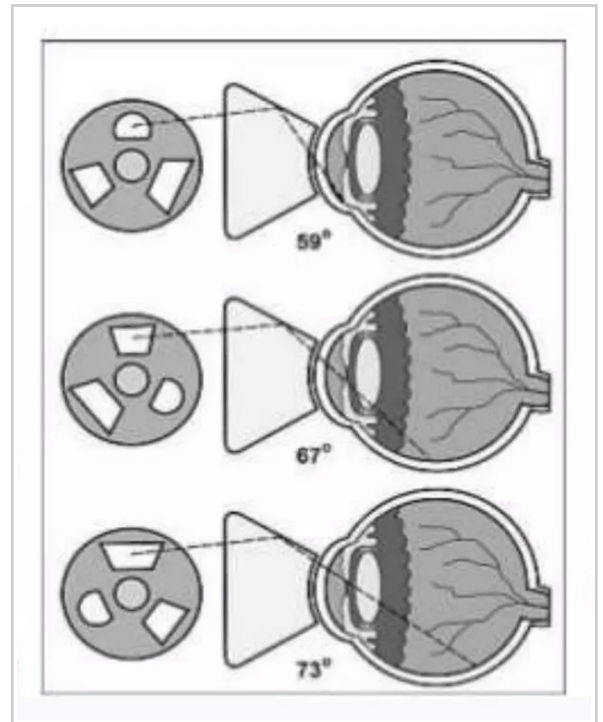
Assessment of severity of thyroid eye disease

NOSPECS

- 0 — **N**o signs or symptoms
- 1 — **O**nly signs, no symptoms
- 2 — **S**oft tissue involvement
- 3 — **P**roptosis
- 4 — **E**xtraocular muscle involvement
- 5 — **C**orneal involvement
- 6 — **S**ight loss

Goldmann three-mirror lens

The smallest the lens, the more peripheral the view



Suspicious eyelid nevus

ABCDE

Asymmetry

Borders irregular

Color mottled

Diameter large

Enlargement over time



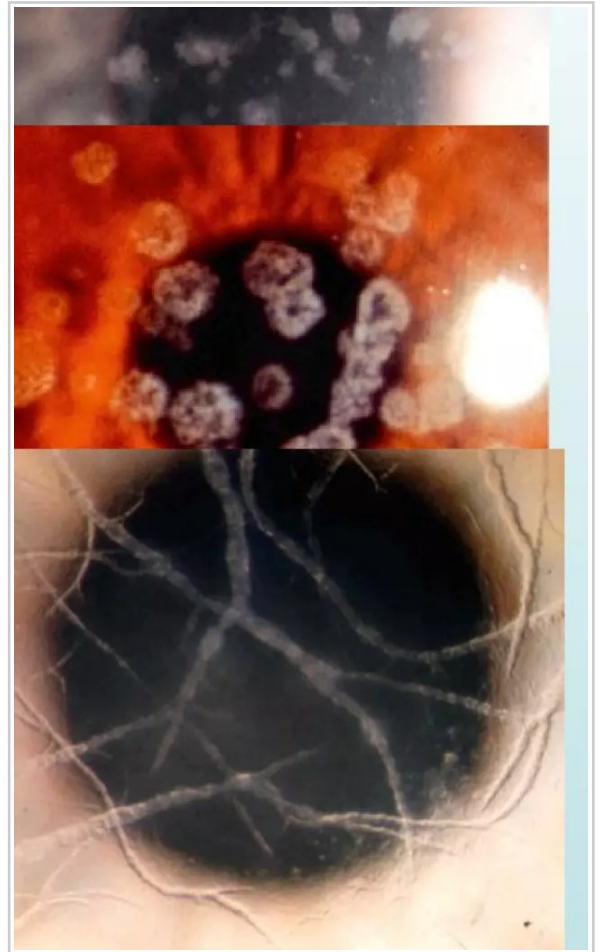
Stromal corneal dystrophies

MARILYN MONROE ALWAYS GETS HER MAN IN LA CITY

Macular dystrophy — **M**ucopolysaccharide — **A**lcian blue

Granular Dystrophy — **H**yaline material — **M**asson's Trichrome

Lattice Dystrophy — **A**myloid — **C**ongo Red



Angle structures

I CAN SEE TILL SCHWALBE'S LINE

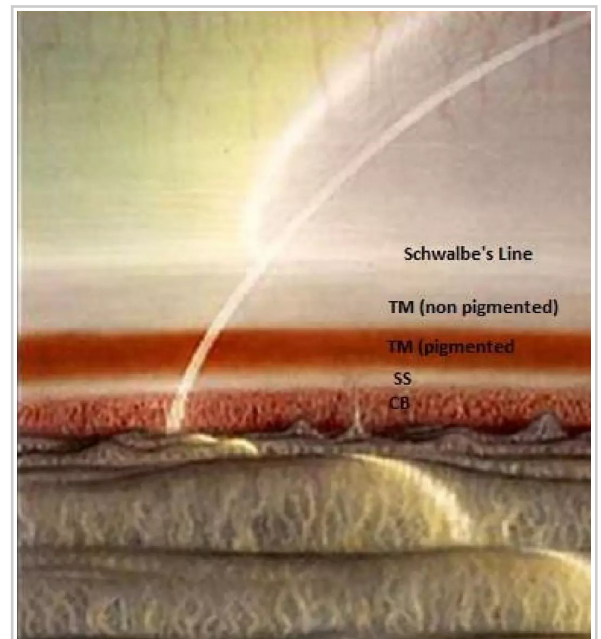
Iris root

Ciliary body

Scleral spur

Trabecular meshwork

Schwalbe's line



Keratoconus

CONES

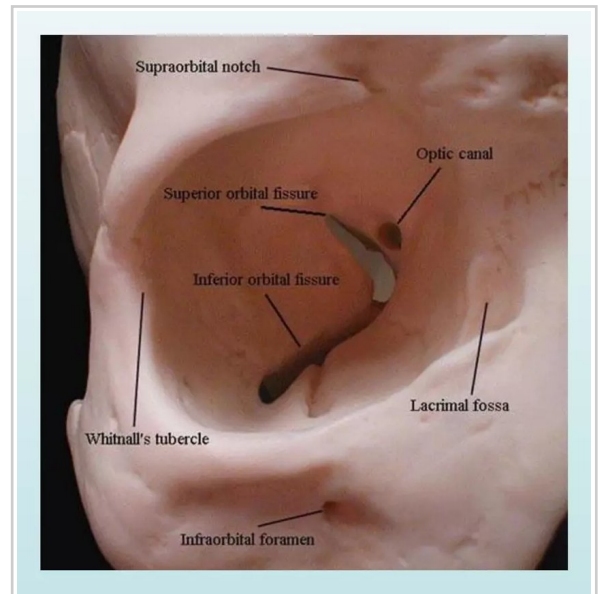
- C**entral scarring & Fleischer ring
- O**il drop reflex / Oedema (hydrops)
- N**erves prominent
- E**xcessive bulging of lower lid on downgaze (Munson's sign)
- S**triae (Vogt's)



Whitnall's tubercle attachments

5 L'S

- L**evator aponeurosis
- L**ateral rectus check ligament
- L**ockwood's ligament
- L**ateral canthal tendon
- L**acrimal gland fascia



A and V patterns

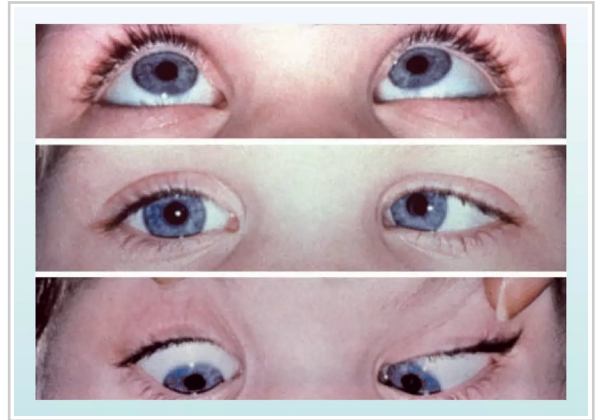
VI/SA

V : V-Pattern

I : IOOA

S : SOOA

A : A-Pattern



Etiology of Neovascular Glaucoma

RUBEOTIC

Retinopathy and Retinal vein occlusion (PDR, CRVO)

Retinal detachment

Uveitis

BRVO

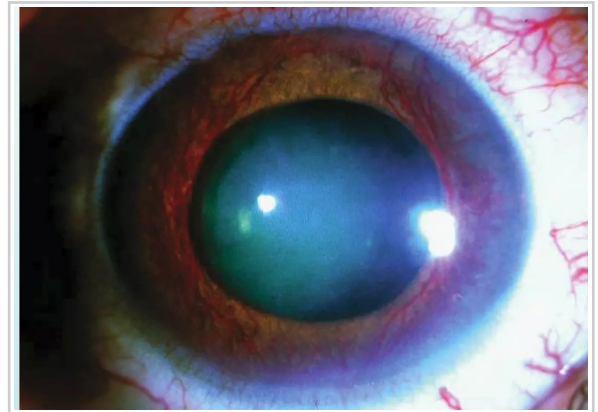
Eale's Disease

Ocular Ischemic Syndrome

Trauma

Intraocular tumours (choroidal melanoma)

Carotid Cavernous Fistula



Argyll Robertson pupil

ARP / PRA

Accommodation Reflex Present

Pupillary Reflex Absent

Indications of vitrectomy

VITRECTOM-Y

- V** — Vitreous hge due to BRVO, PDR, Vasculitis
- I** — Infection – Endophthalmitis
- T** — Traction
- R** — Retinal detachment
- E** — Epiretinal Membrane
- C** — Cataract related complication – nucleus drop, IOL drop
- T** — Trauma causing vit hge, IOFB
- O** — Opacity of vitreous cavity due to severe inflammation not resolved with medical therapy
- M** — Macular disease — macular hole, submacular membrane

Neuroectoderm derivatives

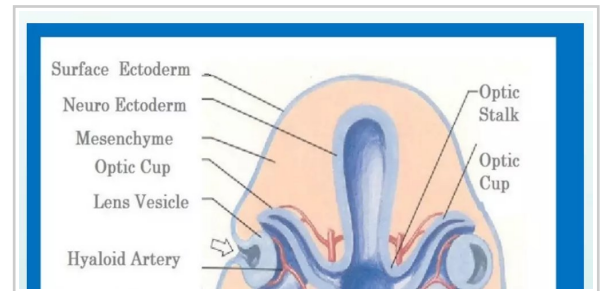
MORE

Muscles of Iris / Pupil

Optic Nerve

Retina

Epithelium of Iris & Ciliary body



Causes of proptosis

VEIN

- Vascular
- Endocrinal
- Inflammatory
- Neoplastic

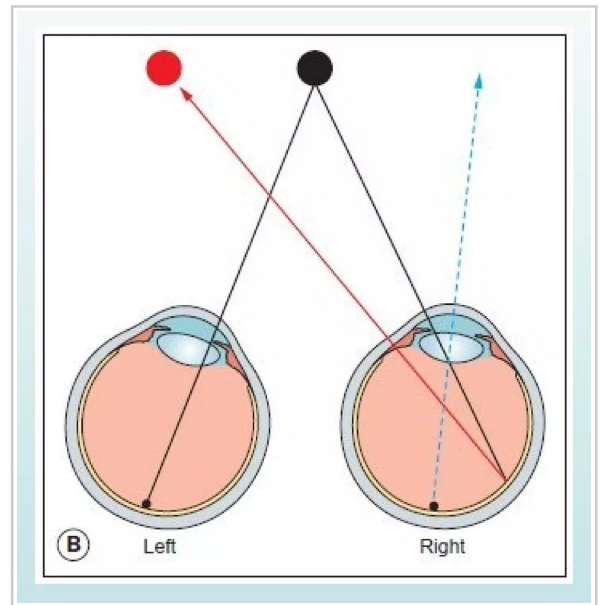


Crossed-uncrossed diplopia

X = CROSSED

eXotropia

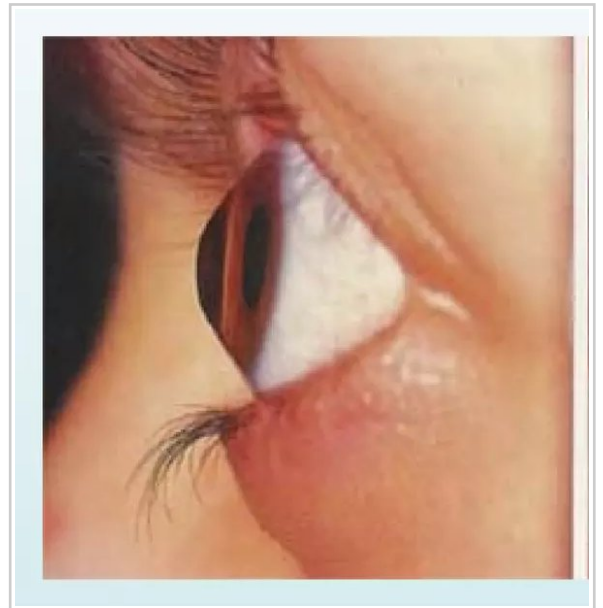
X = Crossed diplopia



Systemic associations of keratoconus

ABCDEF

- A**topy
- B**ones (osteogenesis imperfecta)
- C**rouzon's syndrome
- D**own's syndrome
- E**hler's Danlos syndrome
- F**ingers (Marfan's)



Drugs causing cataract

ABCD

Amiodarone

Busulfan

Chlorpromazine

Dexamethasone

